

- 1 A coffee shop provides free internet access for its customers. It is known that the probability that a randomly selected customer is accessing the internet is 0.35, independently of all other customers.

(i) 10 customers are selected at random.

A Find the probability that exactly 5 of them are accessing the internet.

[3]

B Find the probability that at least 5 of them are accessing the internet.

[2]

C Find the expected number of these customers who are accessing the internet.

[2]

Another coffee shop also provides free internet access. It is suspected that the probability that a randomly selected customer at this coffee shop is accessing the internet may be different from 0.35. A random sample of 20 customers at this coffee shop is selected. Of these, 10 are accessing the internet.

- (ii) Carry out a hypothesis test at the 5% significance level to investigate whether the probability for this coffee shop is different from 0.35. Give a reason for your choice of alternative hypothesis.

[9]

- (iii) To get a more reliable result, a much larger random sample of 200 customers is selected over a period of time, and another hypothesis test is carried out. You are given that 90 of the 200 customers were accessing the internet. You are also given that, if X has the binomial distribution with parameters $n = 200$ and $p = 0.35$, then $P(X \leq 90) = 0.0022$. Using the same hypotheses and significance level which you used in part (ii), complete this test.

[2]

- 2 A researcher is investigating whether people can identify whether a glass of water they are given is bottled water or tap water. She suspects that people do no better than they would by guessing. Twenty people are selected at random; thirteen make a correct identification. She carries out a hypothesis test.

(i) Explain why the null hypothesis should be $p = 0.5$, where p represents the probability that a randomly selected person makes a correct identification.

[2]

(ii) Briefly explain why she uses an alternative hypothesis of $p > 0.5$.

[1]

(iii) Complete the test at the 5% significance level.

[5]

- 3 It is known that on average 85% of seeds of a particular variety of tomato will germinate. Ramesh selects 15 of these seeds at random and sows them.

(i)

A Find the probability that exactly 12 germinate.

[3]

B Find the probability that fewer than 12 germinate.

[2]

The following year Ramesh finds that he still has many seeds left. Because the seeds are now one year old, he suspects that the germination rate will be lower. He conducts a trial by randomly selecting n of these seeds and sowing them. He then carries out a hypothesis test at the 1% significance level to investigate whether he is correct.

(ii) Write down suitable null and alternative hypotheses for the test. Give a reason for your choice of alternative hypothesis.

[4]

(iii) In a trial with $n = 20$, Ramesh finds that 13 seeds germinate. Carry out the test.

[4]

(iv) Suppose instead that Ramesh conducts the trial with $n = 50$, and finds that 33 seeds germinate. Given that the critical value for the test in this case is 35, complete the test.

[3]

(v) If n is small, there is no point in carrying out the test at the 1% significance level, as the null hypothesis cannot be rejected however many seeds germinate. Find the least value of n for which the null hypothesis can be rejected, quoting appropriate probabilities to justify your answer.

[3]

- 4 A drug for treating a particular minor illness cures, on average, 78% of patients. Twenty people with this minor illness are selected at random and treated with the drug.

(i)

A Find the probability that exactly 19 patients are cured.

[3]

B Find the probability that at most 18 patients are cured.

[3]

C Find the expected number of patients who are cured.

[1]

- (ii) A pharmaceutical company is trialling a new drug to treat this illness. Researchers at the company hope that a higher percentage of patients will be cured when given this new drug. Twenty patients are selected at random, and given the new drug. Of these, 19 are cured. Carry out a hypothesis test at the 1% significance level to investigate whether there is any evidence to suggest that the new drug is more effective than the old one.

[8]

- (iii) If the researchers had chosen to carry out the hypothesis test at the 5% significance level, what would the result have been? Justify your answer.

[2]

- 5 To withdraw money from a cash machine, the user has to enter a 4-digit PIN (personal identification number). There are several thousand possible 4-digit PINs, but a survey found that 10% of cash machine users use the PIN '1234'.

(i) 16 cash machine users are selected at random.

(A) Find the probability that exactly 3 of them use 1234 as their PIN.

[3]

(B) Find the probability that at least 3 of them use 1234 as their PIN.

[2]

(C) Find the expected number of them who use 1234 as their PIN.

[1]

An advertising campaign aims to reduce the number of people who use 1234 as their PIN. A hypothesis test is to be carried out to investigate whether the advertising campaign has been successful.

(ii) Write down suitable null and alternative hypotheses for the test. Give a reason for your choice of alternative hypothesis.

[4]

(iii) A random sample of 20 cash machine users is selected.

(A) Explain why the test could not be carried out at the 10% significance level.

[3]

(B) The test is to be carried out at the $k\%$ significance level. State the lowest integer value of k for which the test could result in the rejection of the null hypothesis.

[1]

(iv) A new random sample of 60 cash machine users is selected. It is found that 2 of them use 1234 as their PIN. You are given that, if $X \sim B(60, 0.1)$, then (to 4 decimal places)

$$P(X = 2) = 0.0393,$$

$$P(X < 2) = 0.0138,$$

$$P(X \leq 2) = 0.0530.$$

Using the same hypotheses as in part (ii), carry out the test at the 5% significance level.

[4]

- 6 The scores, X , in Paper 1 of an English examination have an underlying Normal distribution with mean 76 and standard deviation 12. The scores are reported as integer marks. So, for example, a score for which $75.5 \leq X < 76.5$ is reported as 76 marks.

(i) Find the probability that a candidate's reported mark is 76.

[4]

(ii) Find the probability that a candidate's reported mark is at least 80.

[3]

(iii) Three candidates are chosen at random. Find the probability that exactly one of these three candidates' reported marks is at least 80.

[2]

The proportion of candidates who receive an A* grade (the highest grade) must not exceed 10% but should be as close as possible to 10%.

(iv) Find the lowest reported mark that should be awarded an A* grade.

[5]

The scores in Paper 2 of the examination have an underlying Normal distribution with mean μ and standard deviation 12.

(v) Given that 20% of candidates receive a reported mark of 50 or less, find the value of μ .

[4]

- 7 A researcher believes that there may be negative association between the quantity of fertiliser used and the percentage of the population who live in rural areas in different countries. The data below show the percentage of the population who live in rural areas and the fertiliser use measured in kg per hectare, for a random sample of 11 countries.

Percentage of population	33	6	58	35	81	69	61	7	74	71	17
Fertiliser use	76	44	6	68	3	10	7	176	5	137	157

- (i) Draw a scatter diagram to illustrate the data.

[3]

- (ii) Explain why it might not be valid to carry out a test based on the product moment correlation coefficient in this case.

[2]

- (iii) Calculate the value of Spearman's rank correlation coefficient.

[5]

- (iv) Carry out a hypothesis test at the 1% significance level to investigate the researcher's belief.

[6]

- (v) Explain the meaning of '1% significance level'.

[1]

- (vi) In order to carry out a test based on Spearman's rank correlation coefficient, what modelling assumptions, if any, are required about the underlying distribution?

[1]

- 8 A geyser is a hot spring which erupts from time to time. For two geysers, the duration of each eruption, x minutes, and the waiting time until the next eruption, y minutes, are recorded.

- (a) For a random sample of 50 eruptions of the first geyser, the correlation coefficient between x and y is 0.758. The critical value for a 2-tailed hypothesis test for correlation at the 5% level is 0.279. Explain

whether or not there is evidence of correlation in the population of eruptions.

[2]

The scatter diagram in Fig. 9 shows the data from a random sample of 50 eruptions of the second geyser.

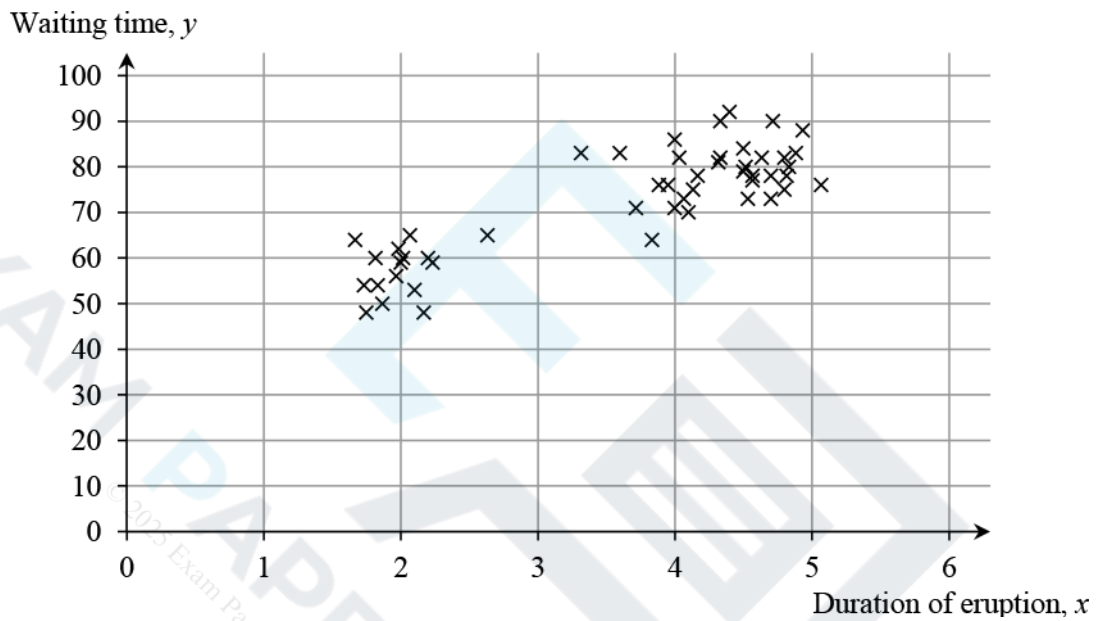


Fig.9

- (b) Stella claims the scatter diagram shows evidence of correlation between duration of eruption and waiting time. Make two comments about Stella's claim.

[2]

- 9 The quality control department of a battery manufacturing company checks the lifetimes of the batteries produced by the company. The lifetimes, x minutes, for a random sample of 80 'Superstrength' batteries are shown in the table below.

Lifetime	$160 \leq x < 165$	$165 \leq x < 168$	$168 \leq x < 170$	$170 \leq x < 172$	$172 \leq x < 175$	$175 \leq x < 180$
Frequency	5	14	20	21	16	4

- (a) Estimate the proportion of these batteries which have a lifetime of at least 174.0 minutes. [2]

- (b) Use the data in the table to estimate

- the sample mean,
- the sample standard deviation.

[3]

The data in the table on the previous page are represented in the following histogram:

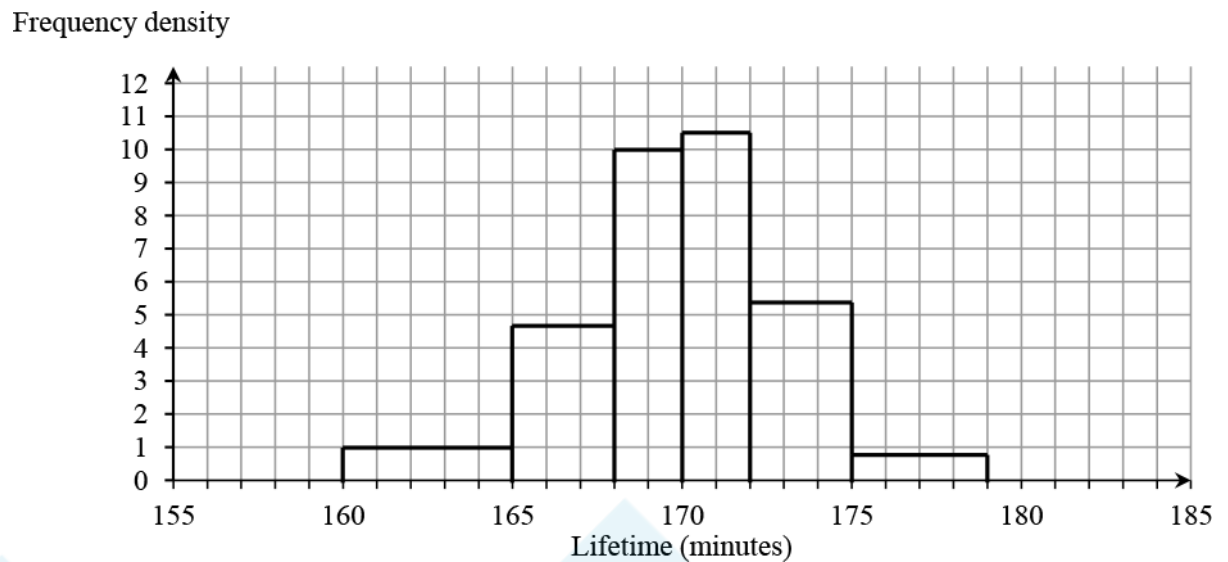


Fig. 15

A quality control manager models the data by a Normal distribution with the mean and standard deviation you calculated in part (b).

(c) Comment briefly on whether the histogram supports this choice of model.

[2]

(d) (i) Use this model to estimate the probability that a randomly selected battery will have a lifetime of more than 174.0 minutes.

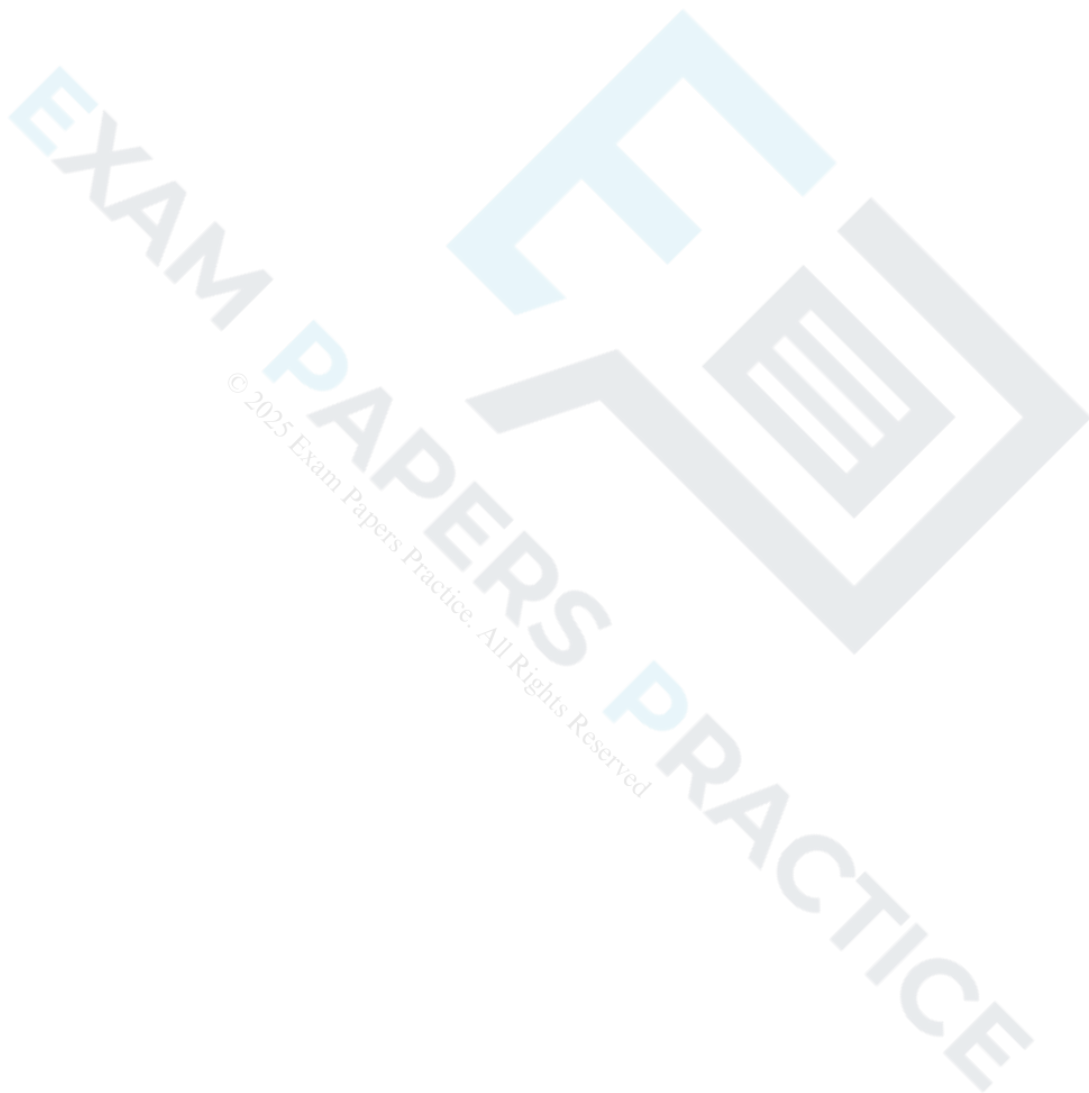
(ii) Compare your answer with your answer to part (a).

[3]

The company also manufactures 'Ultrapower' batteries, which are stated to have a mean lifetime of 210 minutes.

- (e) A random sample of 8 Ultrapower batteries is selected. The mean lifetime of these batteries is 207.3 minutes. Carry out a hypothesis test at the 5% level to investigate whether the mean lifetime is as high as stated. You should use the following hypotheses $H_0 : \mu = 210$, $H_1 : \mu < 210$, where μ represents the population mean for Ultrapower batteries. You should assume that the population is Normally distributed with standard deviation 3.4.

[5]



10 A company operates trains. The company claims that 92% of its trains arrive on time. You should assume that in a random sample of trains, they arrive on time independently of each other.

(a) Assuming that 92% of the company's trains arrive on time, find the probability that in a random sample of 30 trains operated by this company

(i) exactly 28 trains arrive on time,

[2]

(ii) more than 27 trains arrive on time.

[2]

A journalist believes that the percentage of trains operated by this company which arrive on time is lower than 92%.

(b) To investigate the journalist's belief a hypothesis test will be carried out at the 1% significance level. A random sample of 18 trains is selected. For this hypothesis test,

- state the hypotheses,
- find the critical region.

[5]

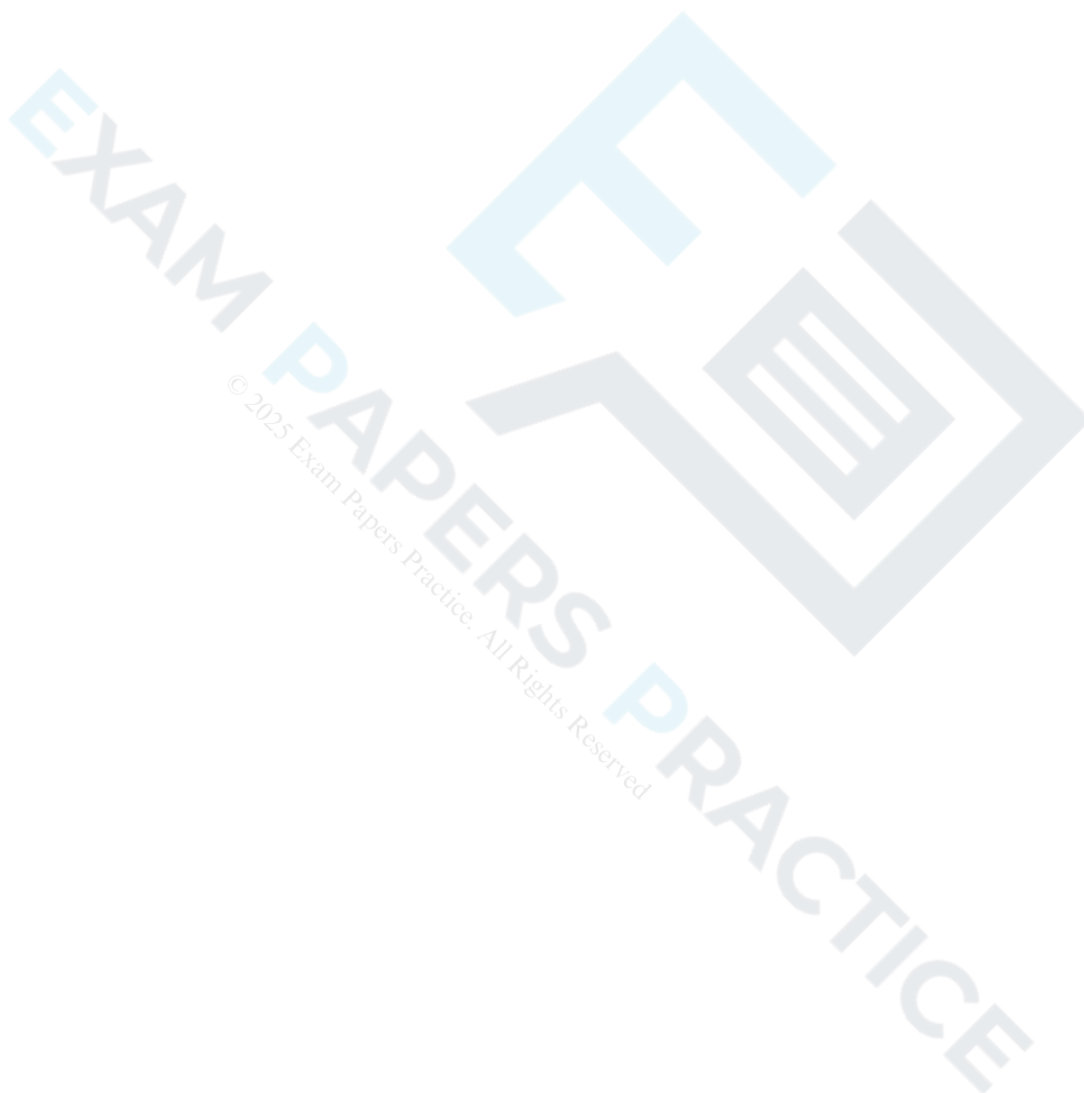
11 In this question you must show detailed reasoning.

EXAM PAPERS PRACTICE

Mr. Evans is standing for re-election to the local council. At the last election 49% of voters voted for Mr. Evans, but it is thought that the level of support for Mr. Evans may have changed. A random sample of 38 voters are asked about their voting intentions and 13 say they intend to vote for Mr. Evans.

Carry out an appropriate hypothesis test at the 5% level to investigate whether or not the level of support for Mr. Evans has changed.

[7]



- 12 The screenshot in Fig. 8 shows the probability distribution for the discrete random variable X , where $X \sim B(20, 0.9)$.

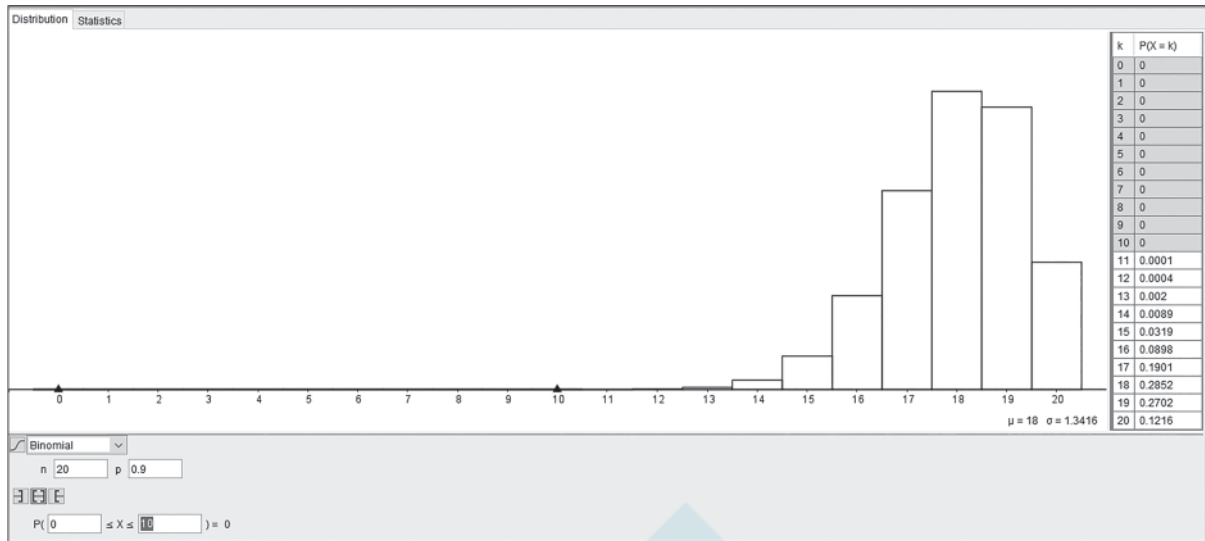


Fig. 8

- (a) Describe the shape of the distribution. [1]
- (b) Explain why the values of $P(X = k)$ for $k = 0$ to 10 inclusive are recorded as 0 in the table in the screenshot. [1]
- (c) State which of the values from 0 to 20 is
- (i) the most likely value of X , [1]
- (ii) the least likely value of X . [1]

- 13 Between the islands of Tenerife and La Gomera there is a resident population of pilot whales. Studies in the past showed that 20% of the adult population were male. Due to a change in environmental conditions, it is thought that the proportion of males has decreased.

In order to investigate this, scientists caught and released a random sample of 43 different adult pilot whales. Exactly 3 of these whales were found to be male.

- (a) Carry out a hypothesis test at the 5% level to investigate whether there is any evidence that the proportion of males has decreased.

[6]

Previous studies also showed that the mean length of adult females was 2.98 metres. On another occasion the scientists caught a random sample of 39 different adult female pilot whales; the length, x metres, of each whale was measured before it was released. The data are summarized below.

$$\sum x = 120.7 \text{ and } \sum x^2 = 384.75$$

- (b) Carry out a hypothesis test at the 5% level to investigate whether there is any evidence that the mean length of adult females has changed.

[8]

14 A type of shampoo is known to relieve the symptoms of 75% of dogs who suffer from a particular minor allergy.

EXAM PAPERS PRACTICE

- (i) 12 dogs who suffer from this allergy are selected at random. Find the probability that the number of these dogs who have their symptoms relieved is
- (A) exactly 9, [3]

(B) at least 9. [2]

A new type of shampoo has been developed to treat the allergy. A hypothesis test is to be carried out to determine whether it relieves the symptoms of a higher proportion of dogs who suffer from the allergy.

- (ii) Write down suitable null and alternative hypotheses for the test. Give a reason for your choice of alternative hypothesis. [4]

A random sample of n dogs who suffer from the allergy is selected.

- (iii) (A) Given that $n = 18$ and the symptoms of 16 dogs are relieved, carry out the test at the 10% significance level.

[4]

- (B) Given instead that $n = 50$ and the symptoms of 42 dogs are relieved, carry out the test at the 10% significance level. You may use the information that, for $X \sim B(50, 0.75)$,

$$P(X = 41) = 0.0721, \quad P(X = 42) = 0.0463, \quad P(X \leq 41) = 0.9084, \quad P(X \leq 42) = 0.9547. \quad [4]$$

- 15 Nigel investigated migration in and out of different areas in the United Kingdom. He considered all the data available for long term international inflow and outflow rates per thousand resident population in 2012. Fig. 9.1 shows a scatter diagram for Nigel's sample after the data had been cleaned.

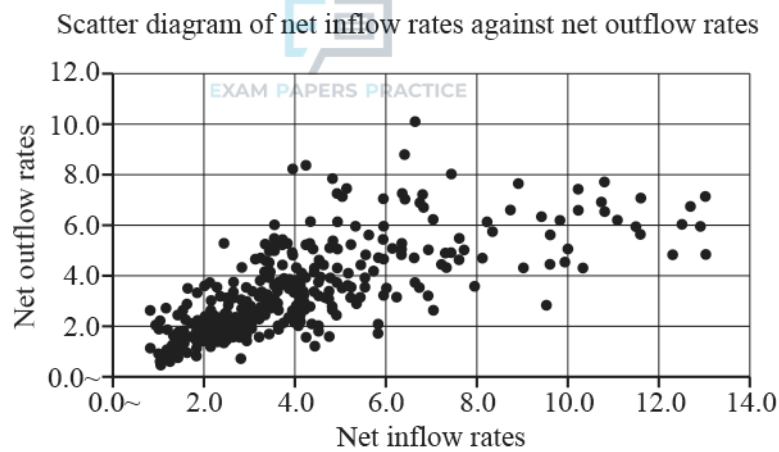


Fig. 9.1

Nigel used software to generate summary statistics. Fig. 9.2 shows summary statistics for inflow rates.

Statistics	
n	391
mean	6.8
s	8.579
$\sum x$	2644.9
$\sum x^2$	46598.17
min	0.8
Q_1	2.5
median	4.0
Q_3	6.8
max	91.8

Fig. 9.2

Nigel decided to clean the data by discarding all outliers, and so the pairs of values with long term inflow rates greater than 13.25 were discarded.

(a) Use information from Fig. 9.2 to show how Nigel calculated the value 13.25.

[1]

Nigel then adopted a similar procedure by considering summary statistics for the long term net outflow rates. As

a result he obtained a sample of size 334 drawn from a population of 391.

(b) State **one** strength and **two** weaknesses of Nigel's method for selecting his sample.

[3]

Nigel found that the product moment correlation coefficient for these data is 0.6916. He used an online statistics calculator to find the associated p -value. A screenshot of the result is shown in Fig. 9.3.

Calculate One, Two Tailed P-Value Correlation Probability

Enter r Value

Enter V Value

t

df

P (One-Tailed)

P (Two-Tailed)

Fig. 9.3

Nigel made the following statement.

“My analysis proves that there is positive correlation between long term net inflow and long term net outflow rates per thousand resident population. Therefore high long term inflow rates cause high long term outflow rates.”

(c) Give **three** reasons why Nigel's statement is flawed.

[3]

- 16 The pre-release data set consists of a subset (see <http://www.ocr.org.uk/Images/308749-units-h630-and-h640-large-data-set-lds-sample-assessment-material.xls>) of the data about countries from the CIA World Facebook.

Data for Croatia shows that the average number of mobile phone subscribers per 1000 population is 1111.72.

(a) Explain how this average can be greater than 1000.

[1]

It is assumed that, once the data have been cleaned, the average number of phones per 1000 population in different countries can reasonably be approximated by a Normal distribution with mean 996 and standard deviation 407.

(b) State two ways in which the data may have been cleaned.

[2]

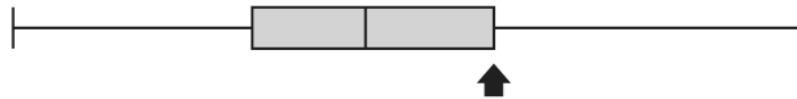


Fig. 11.1

Fig. 11.1 shows a box plot of the cleaned data.

(c) Give the name of the measure which the arrow is pointing to.

[2]

(d) Use the approximate Normal distribution to estimate a value for this measure.

[2]

Fatima thinks that the mean number of mobile phone subscribers per 1000 population for the countries of the world has increased since the data were collected. In order to test this he obtains some up-to-date data for a random sample of countries and uses software to conduct the appropriate hypothesis test at the 5% level of significance. The output from the software is shown in Fig. 11.2.

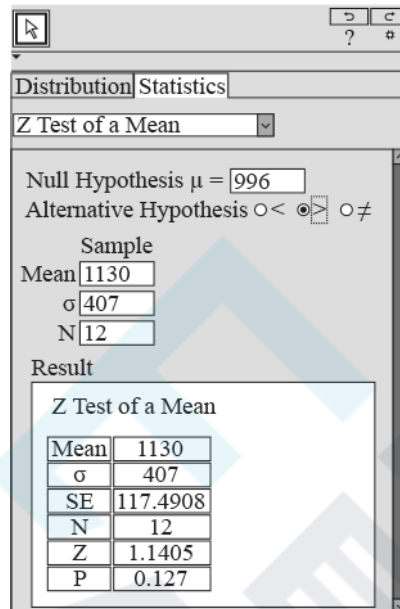


Fig. 11.2

(e) State the conclusion of Fatima's test, explaining your reasoning.

[4]

(f) With reference to the pre-release data set, comment on the sample size Fatima has chosen.

[1]

- 17 A team of biologists are monitoring a subspecies of vole on a remote island. Previous studies have shown that the population mean mass of adult male voles is 26.2 g. It is believed that, due to changes in the environment, the mean mass may have altered. Three of the biologists conducted a new survey and each collected a random sample of adult male voles. The voles were weighed and tagged and then released unharmed.

Fig. 12 shows summary statistics for the 3 samples.

Statistics	Sample 1	Sample 2	Sample 3
Mean	22.6	22.8	23.7
Standard deviation	2.983287	3.258527	4.122196
Variance	8.9	10.618	16.9925
Minimum	19.2	18.6	18.4
Maximum	29.1	30.1	32
Q_1	20.3	20.1	19.8
Median	21.7	22.4	23.2
Q_3	23.95	24.3	26.1
Count	9	11	17

Fig. 12

- (a) For each sample calculate Σx and Σx^2 .

[4]

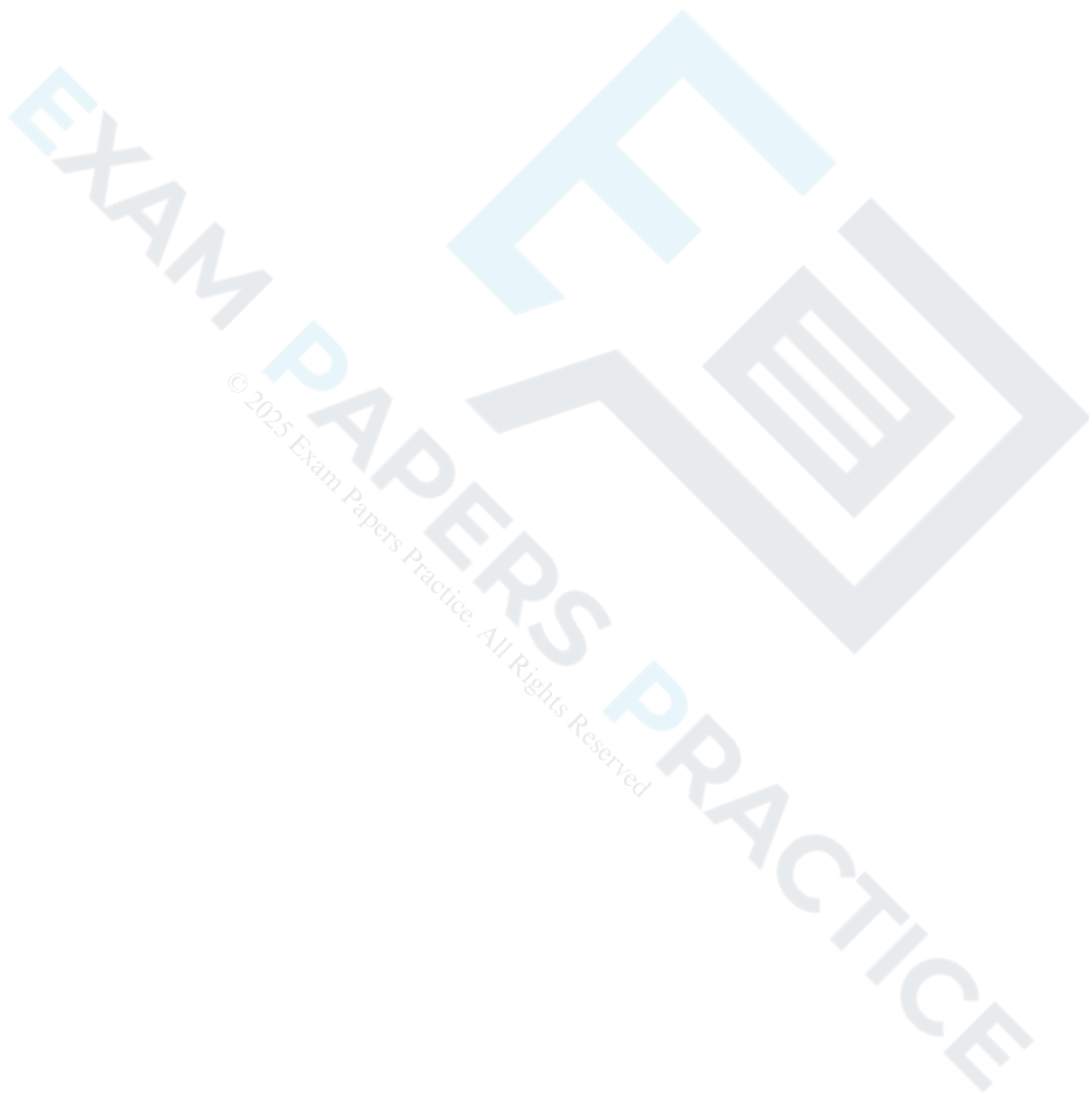
- (b) Use your answers to part (a) to calculate the mean and variance of the mass of all the voles caught in the survey. [2]

The biologists use the results of the survey to test, at the 5% level, the hypothesis $H_0: \mu = 26.1$ against the hypothesis $H_1: \mu \neq 26.1$.

- (c) Assuming that the distribution of the masses of adult male voles may be modelled by the Normal distribution with the value for the variance found in part (b), calculate the critical region for this test. [3]

(d) State the conclusion reached by the biologists, explaining your answer.

[1]



The manufacturers of “Miracleroot” claim that, as long as their instructions are followed, 80% of cuttings taken from Christmas trees will grow roots and develop into healthy new trees. A horticulturist suspects that this claim is optimistic. He takes 500 cuttings, and after carefully following the manufacturer’s instructions, finds that 380 of them developed roots and developed into healthy young trees.

Carry out a hypothesis test at the 1% level to determine whether there is any evidence to suggest that the manufacturer’s claim is optimistic.

[7]



19 In this question you must show detailed reasoning.

EXAM PAPERS PRACTICE

Research showed that in May 2017 62% of adults over 65 years of age in the UK used a certain online social media platform. Later in 2017 it was believed that this proportion had increased. In December 2017 a random sample of 59 adults over 65 years of age in the UK was collected. It was found that 46 of the 59 adults used this online social media platform.

Use a suitable hypothesis test to determine whether there is evidence at the 1% level to suggest that the proportion of adults over 65 in the UK who used this online social media platform had increased from May 2017 to December 2017.

[7]



- 20 Each weekday Keira drives to work with her son Kaito. She always sets off at 8.00 a.m. She models her journey time, x minutes, by the distribution $X \sim N(15, 4)$.

Over a long period of time she notes that her journey takes less than 14 minutes on 7% of the journeys, and takes more than 18 minutes on 31% of the journeys.

- (a) Investigate whether Keira's model is a good fit for the data.

[3]

Kaito believes that Keira's value for the variance is correct, but realises that the mean is not correct.

- (b) Find, correct to two significant figures, the value of the mean that Keira should use in a refined model which does fit the data.

[2]

Keira buys a new car. After driving to work in it each day for several weeks, she randomly selects the journey times for n of these days. Her mean journey time for these n days is 16 minutes. Using the refined model she conducts a hypothesis test to see if her mean journey time has changed, and finds that the result is significant at the 5% level.

(c) Determine the smallest possible value of n .

[5]

- 21 The pre-release material includes data on unemployment rates in different countries. A sample from this material has been taken. All the countries in the sample are in Europe. The data have been grouped and are shown in Fig 14.1.

Unemployment rate	0–	5–	10–	15–	20–	35–50
Frequency	15	21	5	5	2	2

Fig. 14.1

A cumulative frequency curve has been generated for the sample data using a spreadsheet. This is shown in

Fig. 14.2.

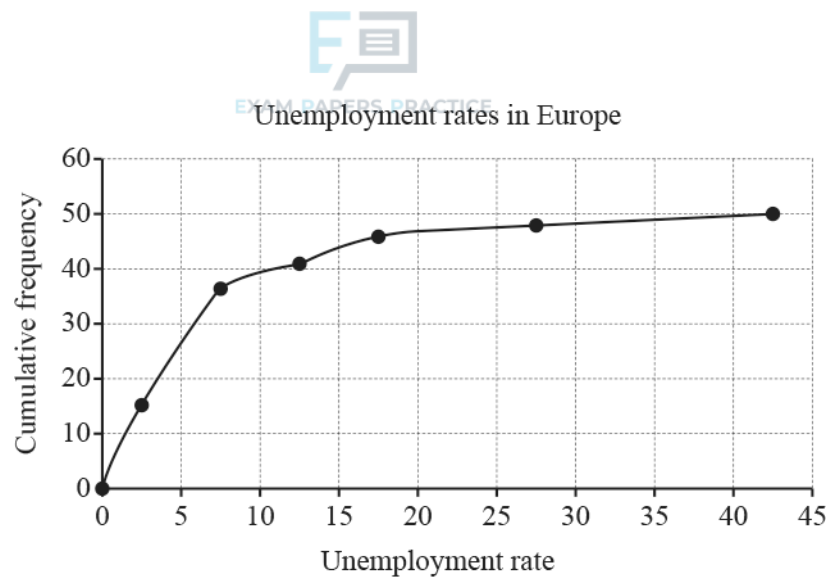


Fig. 14.2

Hodge used Fig. 14.2 to estimate the median unemployment rate in Europe. He obtained the answer 5.0. The correct value for this sample is 6.9.

(a) (i) There is a systematic error in the diagram.

- Identify this error.
- State how this error affects Hodge's estimate.

[2]

(ii) There is another factor which has affected Hodge's estimate.

- Identify this factor.
- State how this factor affects Hodge's estimate.

[2]

- (b) Use your knowledge of the pre-release material to give another reason why any estimation of the median unemployment rate in Europe may be unreliable. [1]
- (c) Use your knowledge of the pre-release material to explain why it is very unlikely that the sample has been randomly selected from the pre-release material. [1]

The scatter diagram shown in Fig. 14.3 shows the unemployment rate and life expectancy at birth for the 47 countries in the sample for which this information is available.

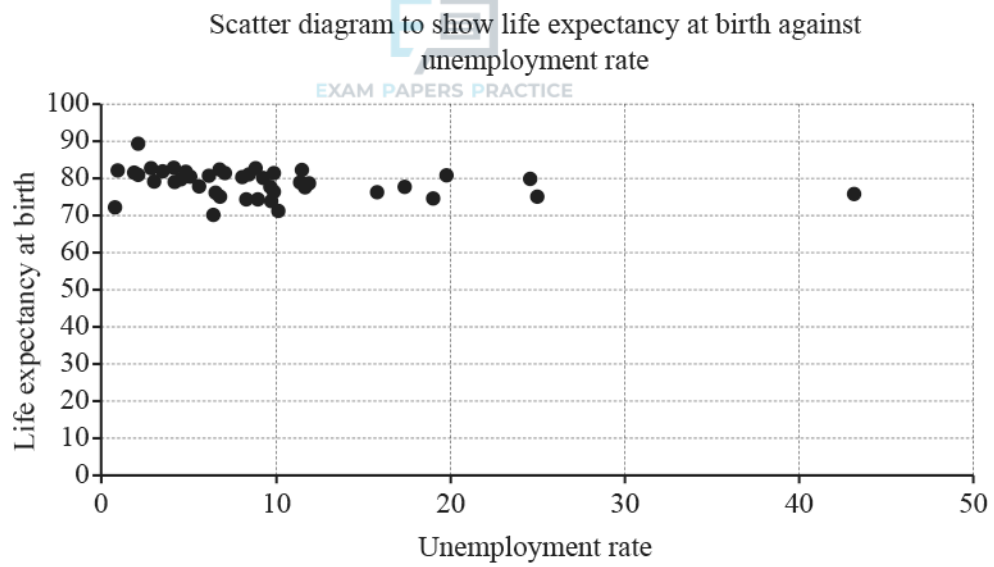


Fig. 14.3

The product moment correlation coefficient for the 47 items in the sample is -0.2607 .

The p -value associated with $r = -0.2607$ and $n = 47$ is 0.0383 .

- (d) Does this information suggest that there is an association between unemployment rate and life expectancy at birth in countries in Europe? [2]

Hodge uses the spreadsheet tools to obtain the equation of a line of best fit for this data.

- (e) The unemployment rate in Kosovo is 35.3 , but there is no data available on life expectancy. Is it reasonable to use Hodge's line of best fit to estimate life expectancy at birth in Kosovo? [1]

Each evening Statto goes for a walk on the same circular route. Over a long period of time Statto noticed that the mean time taken to complete the walk was 56 minutes and the standard deviation was 4 minutes.

A few months ago Statto was ill and was unable to complete the walk for a month. Since then Statto's partner thinks that the mean time Statto takes to complete the evening walk has increased. Over a period of weeks Statto's partner collects a random sample of the times taken by Statto to complete the evening walk.

Statto's partner uses software to generate the summary statistics in Fig. 14.

Statistics ▼	
n	19
Mean	57.4737
σ	3.8848
s	3.9912
Σx	1092
Σx^2	63048
Min	52
Q1	54
Median	57
Q3	60
Max	65

Fig. 14

Use information from Fig. 14 to conduct a hypothesis test to determine whether there is any evidence at the 5% level to suggest that the mean time Statto takes to complete the evening walk has increased. [7]

- 23 Rose packs eggs in boxes of 6, which she then sells at her farm. During the process some eggs are cracked, and Rose randomly selects a sample of boxes and records the number of cracked eggs in each box. The results are summarised in Fig. 16.

Number of cracked eggs	0	1	2	3	4	5	6
Number of boxes	163	103	28	5	0	0	1

Fig.16

- (a) Calculate the mean number of cracked eggs per box.

[1]

Rose believes that the number of cracked eggs per box may be modelled by a binomial distribution.

- (b) State a modelling assumption that is necessary for a binomial distribution to be used to model the number of cracked eggs per box.

[1]

Rose defines p as the probability that a particular egg is cracked.

(c) Use your answer to part (a) to find the value of p .

[2]

(d) Calculate the expected frequencies of cracked eggs per box according to Rose's model, giving your answers correct to 1 decimal place.

[3]

(e) Comment on whether Rose's model is a good fit for the data.

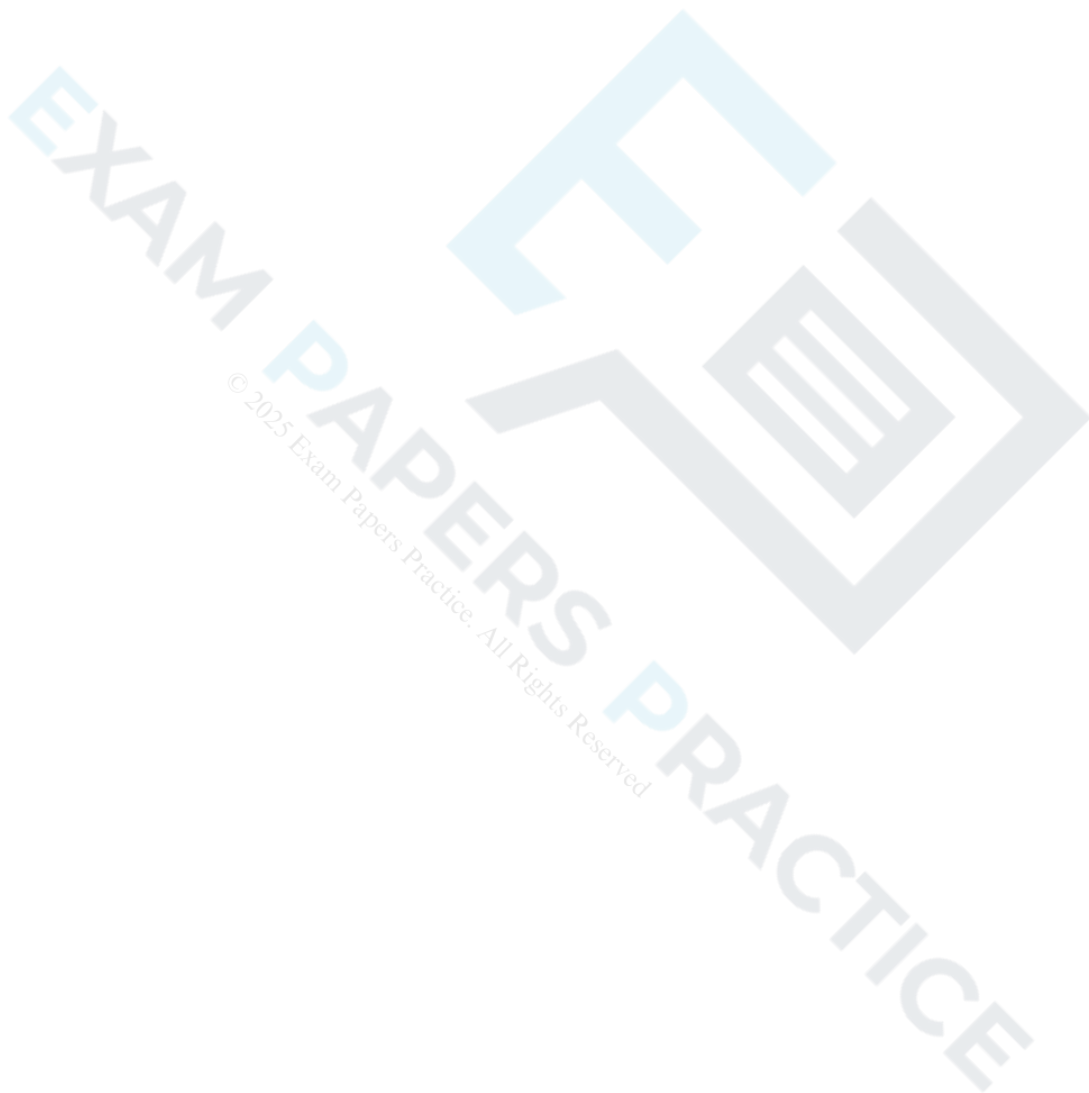
[1]

Instead of selling eggs at the farm, Rose decides to sell them to a wholesaler. The eggs are now selected randomly and packed in open trays of 24. Rose believes that this will result in a change in the probability of an egg being cracked. She selects a tray at random and finds that 5 eggs are cracked.

(f) In this question you must show detailed reasoning.

Conduct a hypothesis test to determine whether there is any evidence at the 5% level to suggest that the proportion of cracked eggs has changed.

[7]



- 24 For a period of 20 years until the end of 2017 the parish council of the village of Springmoor have met every fortnight to discuss local affairs. Throughout this period Mr. Loparts acted as chairperson.

The parish council secretary has noted that the mean length of time of meetings throughout this period was 142 minutes, and the standard deviation of the length of time of meetings was 17.4 minutes.

At the beginning of 2018 Miss Patel took over as chairperson. At the end of 2018 the secretary stated that he believed that the mean length of time of meetings is lower since Miss Patel became chairperson.

The secretary generated a random sample of lengths of times of meetings in 2018. The times, in minutes, are as follows.

123, 137, 119, 135, 136, 143, 126, 126, 125.

Assume that the length of time of meetings may be modelled by the Normal distribution and that the variance of the length of time of meetings is unaltered.

Conduct a hypothesis test at the 5% level to determine whether there is any evidence to support the secretary's belief that the mean length of meeting has decreased.

[7]

25(a) Club 65–80 Holidays fly jets between Liverpool and Magaluf. Over a long period of time records show that half of the flights from Liverpool to Magaluf take less than 153 minutes and 5% of the flights take more than 183 minutes.

An operations manager believes that flight times from Liverpool to Magaluf may be modelled by the Normal distribution.

Use the information above to write down the mean time the operations manager will use in his Normal model for flight times from Liverpool to Magaluf.

[1]

- (b) Use the information above to find the standard deviation the operations manager will use in his Normal model for flight times from Liverpool to Magaluf, giving your answer correct to 1 decimal place.

[3]

- (c) Data is available for 452 flights. A flight time of under 2 hours was recorded in 16 of these flights. Use your answers to parts (a) and (b) to determine whether the model is consistent with this data.

[3]



- (d) The operations manager suspects that the mean time for the journey from Magaluf to Liverpool is less than from Liverpool to Magaluf. He collects a random sample of 24 flight times from Magaluf to Liverpool. He finds that the mean flight time is 143.6 minutes.

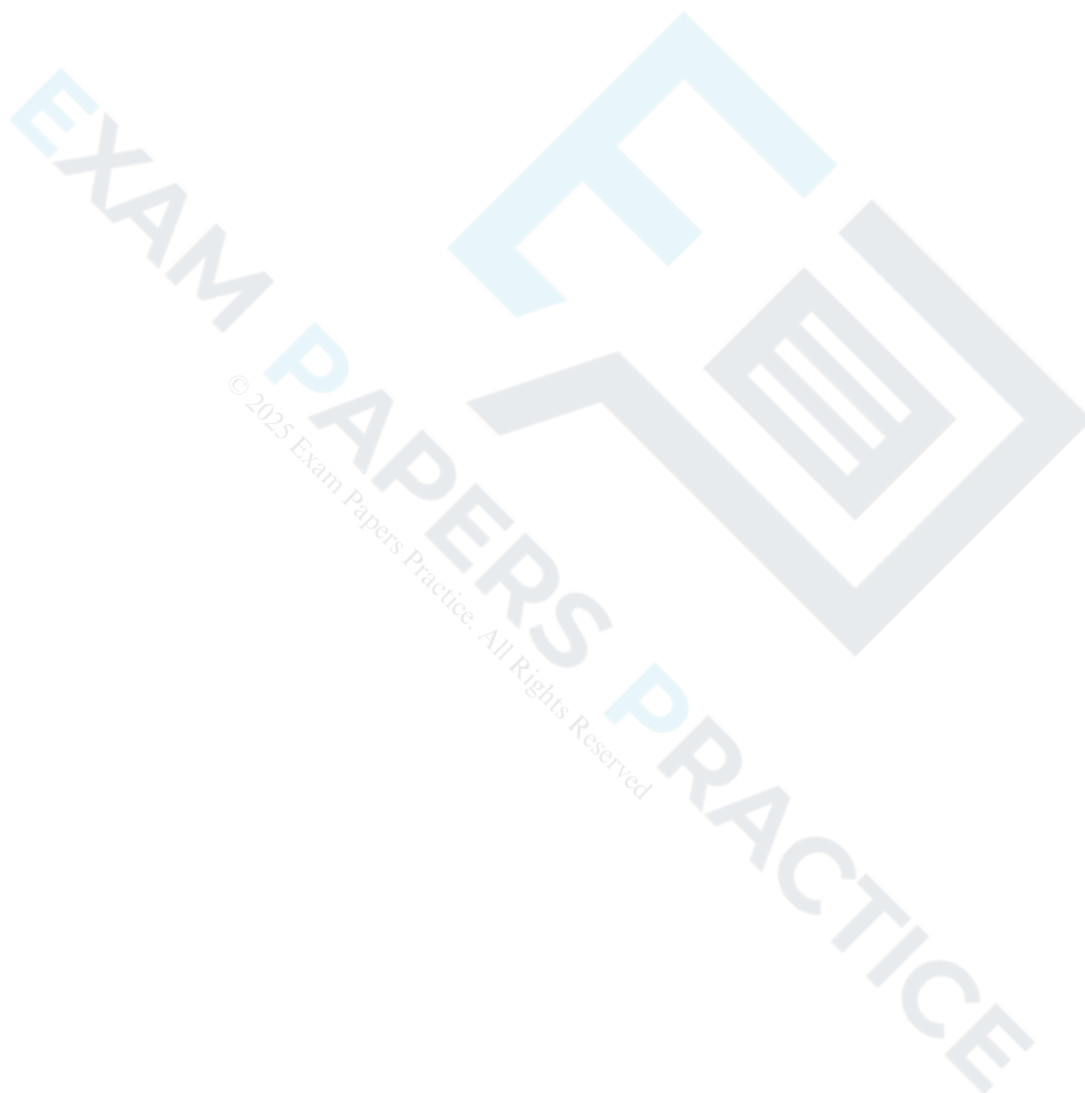
Use the Normal model used in part (c) to conduct a hypothesis test to determine whether there is evidence at the 1% level to suggest that the mean flight time from Magaluf to Liverpool is less than the mean flight time from Liverpool to Magaluf.

[7]



- (e) Identify two ways in which the Normal model for flight times from Liverpool to Magaluf might be adapted to provide a better model for the flight times from Magaluf to Liverpool.

[2]



26(a) The jaguar is a species of big cat native to South America. Records show that 6% of jaguars are born with black coats. Jaguars with black coats are known as black panthers. Due to deforestation a population of jaguars has become isolated in part of the Amazon basin. Researchers believe that the percentage of black panthers may not be 6% in this population.

Find the minimum sample size needed to conduct a two-tailed test to determine whether there is any evidence at the 5% level to suggest that the percentage of black panthers is not 6%.

[3]

- (b) A research team identifies 70 possible sites for monitoring the jaguars remotely. 30 of these sites are randomly selected and cameras are installed. 83 different jaguars are filmed during the evidence gathering period. The team finds that 10 of the jaguars are black panthers.

Conduct a hypothesis test to determine whether the information gathered by the research team provides any evidence at the 5% level to suggest that the percentage of black panthers in this population is not 6%.

[7]

27(a) According to the latest research there are 19.8 million male drivers and 16.2 million female drivers on the roads in the UK.

EXAM PAPERS PRACTICE

A driver in the UK is selected at random. Find the probability that the driver is male.

[1]

(b) Calculate the probability that there are 7 female drivers in a random sample of 25 UK drivers.

[1]



- (c) When driving in a built-up area, Rebecca exceeded the speed limit and was obliged to attend a speed awareness course. Her husband said "It's nearly always male drivers who are speeding." When Rebecca attends the course, she finds that there are 25 drivers, 7 of whom are female. You should assume that the drivers on the speed awareness course constitute a random sample of drivers caught speeding.

In this question you must show detailed reasoning.

Conduct a hypothesis test to determine whether there is any evidence at the 5% level to suggest that male drivers are more likely to exceed the speed limit than female drivers. [7]

(d) State a modelling assumption that is necessary in order to conduct the hypothesis test in part (c).

[1]



28 In a certain country it is known that 11% of people are left-handed.

EXAM PAPERS PRACTICE

An anthropologist believes that the proportion of left-handed people is lower in a particular ethnic group.

The anthropologist collects a random sample of 98 people from this particular ethnic group in order to test the hypothesis that the proportion of left-handed people is less than 11%.

The anthropologist carries out the test at the 1% level.

Determine the critical region for this test.

[3]



- 29 A company supplies computers to businesses. In the past the company has found that computers are kept by businesses for a mean time of 5 years before being replaced. Claud, the manager of the company, thinks that the mean time before replacing computers is now different.

Claud decides to conduct a hypothesis test at the 5% level to test whether there is evidence to suggest that the mean time that businesses keep computers is not 5 years. He takes a random sample of 120 computers. Summary statistics for the length of time computers in this sample are kept are shown in Fig. 9.

Statistics	
n	120
Mean	4.8855
σ	2.6941
s	2.7054
Σx	586.2566
Σx^2	3735.1475
Min	0.1213
Q1	2.5472
Median	4.8692
Q3	7.0349
Max	9.9856

Fig. 9

In this question you must show detailed reasoning.

- State the hypotheses for this test, explaining why the alternative hypothesis takes the form it does.
- Use a suitable distribution to carry out the test.

[8]

30(a) In this question you must show detailed reasoning.

A 5-sided spinner can give scores of 1, 2, 3, 4 or 5. After observing a large number of spins, Elaine models the probability distribution of X , the score on the spinner, as shown in Fig. 12.

x	1	2	3	4	5
$P(X = x)$	0.2	0.3	p	p	q

Fig. 12

When the spinner is spun twice, the probability of obtaining a total score of 9 is 0.06.

Given that $q < 2p$, determine the values of p and q .

[6]

(b) The spinner is spun 10 times. Calculate the probability that exactly one 5 is obtained.

[2]

- (c) Elaine's teacher believes that the probability that the spinner shows a 1 is greater than 0.2. The spinner is spun 100 times and gives a score of 1 on 28 occasions.

Conduct a hypothesis test at the 5% level to determine whether there is any evidence to suggest that the probability of obtaining a score of 1 is greater than 0.2. [7]

31(a) The pre-release material contains information concerning median house prices, recycling rates and employment rates. Fig. 13.1 shows a scatter diagram of recycling rate against employment rate for a random sample of 33 regions.

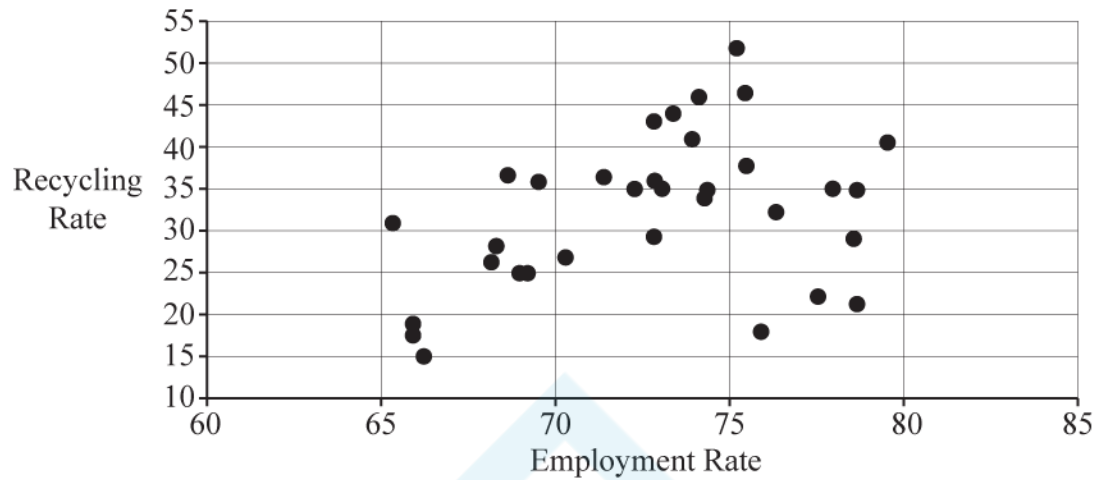


Fig. 13.1

The product moment correlation coefficient for this sample is 0.37154 and the associated p -value is 0.033.

Lee conducts a hypothesis test at the 5% level to test whether there is any evidence to suggest there is positive correlation between recycling rate and employment rate. He concludes that there is no evidence to suggest positive correlation because $0.033 \approx 0$ and $0.37154 > 0.05$.

Explain whether Lee's reasoning is correct.

[2]

- (b) Fig. 13.2 shows a scatter diagram of recycling rate against median house price for a random sample of 33 regions.

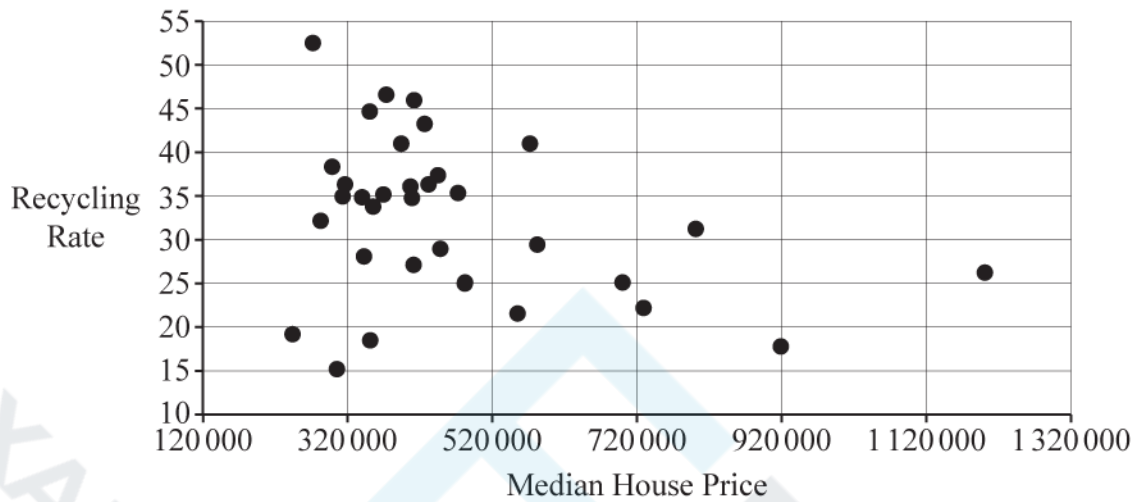


Fig. 13.2

The product moment correlation coefficient for this sample is -0.33278 and the associated p -value is 0.058 .

Fig. 13.3 shows summary statistics for the median house prices for the data in this sample.

Statistics	
n	33
Mean	465467.9697
σ	201236.1345
s	204356.2606
Σx	15360443
Σx^2	8486161617387
Min	243500
Q1	342500
Median	410000
Q3	521000
Max	1200000

Fig. 13.3

Use the information in Fig. 13.3 and Fig. 13.2 to show that there are at least two outliers.

[2]

(c) Describe the effect of removing the outliers on

- the product moment correlation coefficient between recycling rate and median house price,
- the p -value associated with this correlation coefficient, in each case explaining your answer.

[2]

(d) All 33 items in the sample are areas in London. A student suggests that it is very unlikely that only areas in London would be selected in a random sample.

Use your knowledge of the pre-release material to explain whether you think the student's suggestion is reasonable.

[1]

END OF QUESTION PAPER